Stat 502 Project

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Class Session A

Project Task

Your assigned project using dataset(s) in the general topic of **Movies and Videos** can be found in the

TMBD-movie-metadata folder at the following link:

https://drive.google.com/drive/folders/1oHUhSoEf3mYbLdqM4-9wK6GUQgO6oMtU

Please refer to the file **README-TMBD-movie-metadata.txt** in that folder.

Here is your assigned task for the term project:

- Core Task, to be completed by the Preliminary Review: Given a genre, report the relative frequency of movies for which a producer, executive producer or director of a given movie also appeared as a cast member in that movie (as listed in the credits).
- Final Project Submission, due in Finals Week: Your program should provide a SAS macro that allows the user to specify a list of genres to be compared.

Abstract

This project analyzes the frequency with which movie producers, executive producers, and directors also appear as cast members in their own films, examining variations across different movie genres. Using the TMDB 5000 Movies dataset, the analysis reveals significant differences in overlap frequencies: Animation films show the highest rate of 29.06%, followed by Documentaries, 23.64% and Comedies 23.14%, while Foreign films show the lowest of 5.88%. A SAS macro was developed allowing users to compare any combination of genres, with case-insensitive input handling and formatted percentage outputs.

Introduction

This analysis examines how frequently movie producers, executive producers, and directors also appear as cast members in that movie. The dataset contains over 4,000 movies from The Movie Database (TMDB).

Results and Analysis

The tables below shows the resulta of the analysis.

Table 1: Relative Frequency of Producer/Director Cast Overlap by Genre

genre	total_movies_in_genre	movies_with_overlap	relative_frequency
Animation	234	68	29.06%
Documentary	110	26	23.64%
Comedy	1720	398	23.14%
Western	79	17	21.52%
Music	183	34	18.58%
Crime	694	125	18.01%
Family	503	88	17.50%
Romance	890	153	17.19%
Science Fiction	503	81	16.10%
Adventure	784	126	16.07%
Drama	2297	369	16.06%
Action	1152	182	15.80%
Thriller	1248	172	13.78%
War	142	19	13.38%
Horror	517	68	13.15%
TV Movie	8	1	12.50%
Mystery	347	41	11.82%
Fantasy	418	49	11.72%
History	197	23	11.68%
Foreign	34	2	5.88%

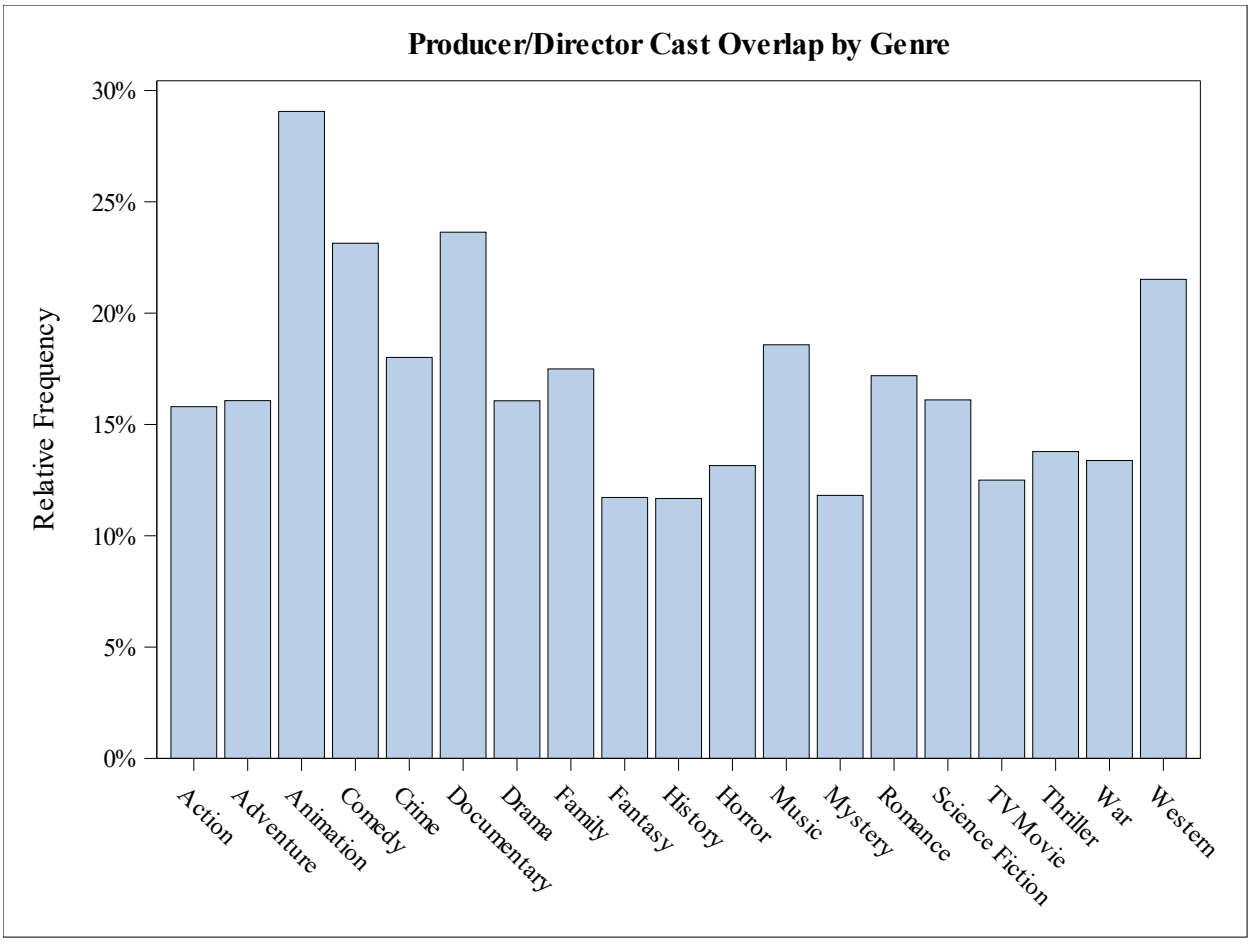


Figure 1: Bar Chart of Producer Cast Overlap by Genre

The bar chart visualizes genres with overlap frequencies exceeding 10%, clearly showing Animation and Documentary leading in producer/director cast participation.

Method

1. The analysis involved parsing JSON data from the TMDB dataset to extract
- Movie genres
- Cast members and their characters
- Crew members focusing on producers and directors
2. SQL joins identified overlaps where producers and directors also appeared in cast roles, and relative frequencies were calculated by genre.

Discussion

Table 1 and Figure 1 shows differences in how often producers or directors also appear as cast in movies genres. Animation films have the highest overlap of about 29.06%. Documentaries and comedies also show high overlap with approximately 24% and 23.14% likely because filmmakers and writers often appear in their own productions. Moderate overlap rates are seen in Westerns, Music, and Crime films with 21.52%, 18.58%, and 18.01% respectively, where some creators participate on-screen. The lowest overlap occurs in Foreign films with 5.88% followed by History with 11.68% and Fantasy, 11.72%, which may reflect larger production scales, language differences, or the need for specialized actors in these genres.

Macros

- Flexible Analysis: Allows comparison of producer/director cast overlap across any combination of genres
- Case-Insensitive: Users can input genres in any capitalization (e.g., 'action', 'ACTION', 'Action')
- Formatted Output: Produces clean tables with percentage formatting

Conclusion

This analysis reveals the fascinating intersection of creative roles in filmmaking, showing how genre conventions influence whether producers and directors step in front of the camera. The developed macro provides researchers and industry professionals with a valuable tool for exploring these patterns across any combination of film genres.

SAS Codes

```
/*
Program:    Final_Project.sas
Author:     Abigail Amoah
Date:       12/09/2025
Purpose:    Analyze the frequency of producer/director cast overlap
                  across different movie genres in the TMDB 5000 dataset.

/*
1. I placed this SAS file in the same folder containing:
      - tmdb_5000_movies.csv
      - tmdb_5000_credits.csv.zip
2. Modify the folder path in the %LET statement below to point to
the folder containing the data files.

3. Run the ENTIRE SAS file to:
 - Import and process the datasets
 - Define the analysis macro

4. For the macro,
      MACRO: AbiMacro
      PURPOSE: Analyze producer/director cast overlap for specified genres
    PARAMETERS: genres - One or more genre names in single quotes, separated
by commas
    OUTPUT: Prints table showing overlap statistics for specified genres

5. After running the entire file, open a new SAS Editor window
and invoke the macro with desired genres using:

Example for checking if macro works
%AbiMacro('Action');
%AbiMacro(%str('Animation','Documentary'));
%%AbiMacro(%str('action','Comedy','Drama'));

Genre names are case-insensitive */

/* SET FOLDER PATH */

%let folder = C:\Users\abiga\Desktop\STA 502\PROJECT;

/* STEP 1: DATA IMPORT AND PREPARATION */
```

```
proc import datafile="&folder\tmdb_5000_movies.csv"
    out=movies
    dbms=csv replace;
    getnames=yes;
run;

/* Credits from ZIP archive */
filename credits ZIP "&folder\tmdb_5000_credits.csv.zip"
    member="tmdb_5000_credits.csv";

data credits;
    length movie_id 8
           title $800
           cast $32767
           crew $32767;
    infile credits dsd;
    input movie_id title cast crew;
run;

/* Keep only needed variables */
data tmdb_5000_movies;
    set movies;
    keep id title genres;
run;

data tmdb_5000_credits;
    set credits;
    keep movie_id cast crew;
run;

/* Parse genres from JSON format */
data movie_genres;
    set tmdb_5000_movies;
    length genre $50;
    keep id title genre;

    if not missing(genres) then do;
        work_string = genres;
        do while(index(work_string, '"name": "') > 0);
            name_pos = index(work_string, '"name": "');
            if name_pos > 0 then do;
                text_start = name_pos + 9;
                text_end = index(substr(work_string, text_start), '"') +
text_start - 2;
                if text_end >= text_start then do;
                    genre = substr(work_string, text_start, text_end -
text_start + 1);
                    if not missing(genre) then output;
                    work_string = substr(work_string, text_end + 2);
                end;
                else leave;
            end;
            else leave;
        end;
    end;
    drop work_string name_pos text_start text_end;
run;
```

```
/* Parse cast members */
data movie_cast;
    set tmdb_5000_credits;
    length cast_member_name $100 character $200;
    keep movie_id cast_member_name character;

    temp_cast = compress(cast, '[] ');
    cast_count = countw(temp_cast, '{}');

    do i = 1 to cast_count;
        cast_item = scan(temp_cast, i, '{}');
        if index(cast_item, 'name') > 0 then do;
            name_pos = index(cast_item, '"name":');
            if name_pos > 0 then do;
                name_start = name_pos + 8;
                name_end = index(substr(cast_item, name_start), '"') +
name_start - 2;
                cast_member_name = substr(cast_item, name_start, name_end -
name_start + 1);

                char_pos = index(cast_item, '"character":');
                if char_pos > 0 then do;
                    char_start = char_pos + 13;
                    char_end = index(substr(cast_item, char_start), '"') +
char_start - 2;
                    character = substr(cast_item, char_start, char_end -
char_start + 1);
                end;

                output;
            end;
        end;
    end;
    drop temp_cast cast_count i cast_item name_pos name_start name_end
         char_pos char_start char_end;
run;

/* Parse crew members (producers and directors only) */
data movie_crew;
    set tmdb_5000_credits;
    length crew_member_name $100 job $50;
    keep movie_id crew_member_name job;

    temp_crew = compress(crew, '[] ');
    crew_count = countw(temp_crew, '{}');

    do i = 1 to crew_count;
        crew_item = scan(temp_crew, i, '{}');
        if index(crew_item, 'name') > 0 then do;
            name_pos = index(crew_item, '"name":');
            job_pos = index(crew_item, '"job":');

            if name_pos > 0 and job_pos > 0 then do;
                name_start = name_pos + 8;
                name_end = index(substr(crew_item, name_start), '"') +
name_start - 2;
```

```
                crew_member_name = substr(crew_item, name_start, name_end -
name_start + 1);

                job_start = job_pos + 7;
                job_end = index(substr(crew_item, job_start), '"') +
job_start - 2;
                job = substr(crew_item, job_start, job_end - job_start + 1);

                if upcase(job) in ('PRODUCER', 'EXECUTIVE PRODUCER',
'DIRECTOR') then output;
            end;
        end;
    end;
    drop temp_crew crew_count i crew_item name_pos name_start name_end
         job_pos job_start job_end;
run;

/* Find movies where producers/directors also appear in cast */
proc sql;
    create table overlap_movies as
    select distinct
        c.movie_id,
        c.crew_member_name as person_name,
        c.job,
        ca.character
    from movie_crew c
    inner join movie_cast ca on c.movie_id = ca.movie_id
                            and c.crew_member_name = ca.cast_member_name;
quit;

/* Create full genre overlap statistics for reference */
proc sql;
    create table genre_overlap_stats as
    select
        mg.genre,
        count(distinct mg.id) as total_movies_in_genre,
        count(distinct case when om.movie_id is not null then mg.id end) as
movies_with_overlap,
        calculated movies_with_overlap / calculated total_movies_in_genre as
relative_frequency
    from movie_genres mg
    left join overlap_movies om on mg.id = om.movie_id
    group by mg.genre
    order by relative_frequency desc;
quit;

/*
/*rtf output
ods rtf file="C:\Users\abiga\Desktop\STA 502\PROJECT\movie frequency.rtf";

proc print data=genre_overlap_stats noobs;
    title "Relative Frequency of Producer/Director Cast Overlap by Genre";
    var genre total_movies_in_genre movies_with_overlap relative_frequency;
    format relative_frequency percent8.2;
run;

/* Bar chart
```

```
proc sgplot data=genre_overlap_stats;
    where relative_frequency > 0.10;
    vbar genre / response=relative_frequency;
    yaxis label="Relative Frequency" valuesformat=percent8.0;
    title "Producer/Director Cast Overlap by Genre";
    xaxis display=(nolabel) fitpolicy=rotate;
run;
ods rtf close;
*/

/* Defining Macros */

%macro AbiMacro(genres);
    /* Process specified genres */
    data selected_genres;
        set movie_genres;
        where upcase(genre) in (%upcase(&genres));
    run;

    proc sql;
        create table results as
        select
            sg.genre,
            count(distinct sg.id) as total_movies,
            count(distinct case when om.movie_id is not null then sg.id end)
as overlap_movies,
            calculated overlap_movies / calculated total_movies as
relative_frequency format=percent8.2
        from selected_genres sg
        left join overlap_movies om on sg.id = om.movie_id
        group by sg.genre
        order by relative_frequency desc;
    quit;

    /* Display results */
    title "Relative Frequency of Producer/Director Cast Overlap";
    proc print data=results noobs;
        var genre total_movies overlap_movies relative_frequency;
    run;
    title;
%mend AbiMacro;

/* Testing the Macro

%AbiMacro('Action');

%AbiMacro(%str('Animation','Documentary'));

%AbiMacro(%str('action','Comedy','Drama'));

*/
```